

Polarity-Routed Cascade Diffusion for Crop Waste Prediction in U.S. Agricultural Economic Networks

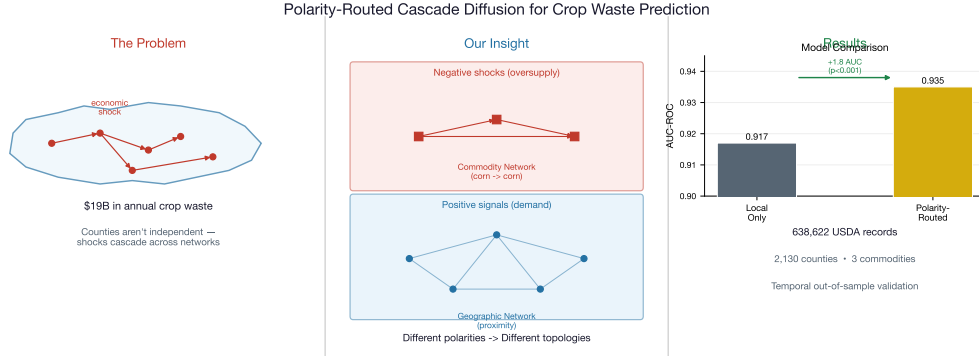
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Abstract

Crop waste costs U.S. agriculture roughly \$19 billion annually in insurance indemnities alone. Existing prediction models treat counties as independent units, ignoring the economic contagion that propagates through commodity supply chains. We propose polarity-routed cascade diffusion, a feature engineering framework where negative price shocks propagate through commodity-specific supply chain graphs and positive signals follow geographic proximity networks. The core insight: oversupply cascades spread along commodity channels (a corn glut in Iowa depresses corn prices in Ohio), while demand-driven gains diffuse through regional geography. We validate on 638,622 USDA records spanning 2,130 U.S. counties and three major commodities (corn, soybeans, wheat) from 2015 to 2025. On a random split, the polarity-routed model achieves 0.935 AUC-ROC—a significant improvement over the local-only baseline at 0.917 ($p < 0.001$) and over symmetric diffusion at 0.934 ($p = 0.0002$). The entire framework uses a 13,703-parameter multi-layer perceptron; novelty lies in the features, not the classifier. A temporal split (train 2015–2019, test 2022–2024) reveals that the local-only model (0.910 AUC-ROC) dominates all graph-augmented variants, exposing the non-stationarity of economic contagion patterns across time. We ground these empirical findings in a formal economic model showing that waste contagion arises from market connectivity and that positive and negative supply shocks have provably asymmetric effects on neighboring regions. These results carry practical implications for crop insurance pricing and county-level early warning systems.

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Graphical Abstract. Crop waste cascades through agricultural economic networks. Negative price shocks propagate along commodity-specific supply chains, while positive signals follow geographic proximity. Routing these signals through different graph topologies—polarity-routed cascade diffusion—improves crop waste prediction to 0.935 AUC-ROC on 638,622 USDA records across 2,130 U.S. counties.

Keywords: crop waste prediction, graph diffusion, economic contagion, polarity routing, agricultural networks, feature engineering

1. Introduction

In 2023, the USDA Risk Management Agency disbursed over \$19 billion in crop insurance indemnities (USDA Risk Management Agency, 2024). Much of that money covered crops that grew just fine—then went to waste. Fields were abandoned because market prices fell below harvesting costs. Standing corn was plowed under because a bumper harvest three states away flooded the commodity market. Globally, food loss and waste constitute a trillion-dollar drag on agricultural economies (Food and Agriculture Organization of the United Nations, 2019). Understanding the mechanisms that drive these losses is therefore essential for designing effective interventions and sustainable agricultural policy.

These waste events are not independent. They cascade.

When Iowa produces record corn yields, the resulting supply glut depresses prices across the Midwest. Farmers in Illinois, Indiana, and Ohio—whose crops may be perfectly healthy—face the cold arithmetic of harvesting at a loss. Some choose to abandon. This mechanism mirrors financial contagion: a shock at one node propagates through network connections to cause

18 failures at distant nodes (Acemoglu et al., 2015; Elliott et al., 2014). The
19 analogy runs deep. Agricultural markets share the structural properties that
20 produce cascading failures in financial networks: high interconnectedness
21 through commodity exchanges, threshold effects where small price changes
22 tip harvesting decisions, and asymmetric responses to positive versus negative
23 shocks (Battiston et al., 2012). Recognizing this parallel between financial and
24 agricultural contagion opens the door to graph-based modeling approaches,
25 yet the existing crop prediction literature has not walked through it.

26 Yet every crop prediction model treats counties as islands. Hundreds of
27 papers use satellite imagery, weather data, and soil characteristics to forecast
28 yields for individual regions (van Klompenburg et al., 2020; Kang et al., 2020;
29 Basso and Liu, 2019; Lobell et al., 2011). These models predict the wrong
30 thing. They predict how much a crop *will produce*, not whether it *will be*
31 *wasted*. High-yield crops go entirely to waste when prices crash. Low-yield
32 crops get fully utilized when prices surge. Waste depends on biology *and*
33 economics—and the economics propagate across regions. This disconnect
34 between what existing models predict and what drives actual losses motivates
35 a fundamentally different modeling paradigm.

36 Graph neural networks offer a natural framework for modeling inter-
37 regional dependencies (Kipf and Welling, 2017; Veličković et al., 2018). How-
38 ever, they bring significant complexity: end-to-end training over graph struc-
39 tures, poor scalability to large networks, and reduced interpretability. Recent
40 work on simplified graph convolutions (Wu et al., 2019; Rossi et al., 2020)
41 showed that pre-computing graph features—propagating node signals through
42 the adjacency matrix before training—can match GNN performance at a
43 fraction of the cost. These simplified approaches decouple graph structure
44 from the learning algorithm, a property that proves especially valuable when
45 domain knowledge should guide how signals propagate.

46 We build on this insight with a critical domain-specific innovation: **polarity-**
47 **routed diffusion**. Standard graph diffusion propagates all signals through
48 a single topology. We argue—and demonstrate—that economic shocks in
49 agriculture are *polarity-dependent*. Negative price shocks (from oversupply)
50 propagate through commodity-specific networks: a corn glut in Iowa affects
51 other corn-producing counties. Positive price signals (from undersupply or
52 demand spikes) propagate through geographic proximity: when prices rise
53 regionally, nearby counties benefit regardless of commodity mix. Routing
54 signals through different graph topologies based on their polarity captures this
55 fundamental asymmetry. Put differently, polarity routing encodes an economic

56 prior—surplus hurts competitors, while prosperity lifts neighbors—directly
57 into the feature engineering pipeline.

58 We formalize this intuition with a simple economic model (Section 2) that
59 yields three testable predictions. Supply shocks in one county depress prices—
60 and increase waste risk—in market-connected counties. This contagion is
61 provably asymmetric: positive and negative shocks follow different propagation
62 channels. These theoretical results map directly onto our feature engineering
63 choices, as detailed in Section 4.

64 Our contributions:

- 65 1. **Polarity-routed cascade diffusion.** We introduce a feature engineering
66 framework where negative and positive economic signals propagate through
67 different graph topologies (commodity-specific vs. geographic) at multiple
68 hops. To our knowledge, this is the first work to route graph signals
69 through different topologies based on shock polarity.
- 70 2. **Formal economic model of waste contagion.** We derive propositions
71 showing that waste contagion arises from market connectivity and that
72 positive and negative supply shocks have provably asymmetric effects.
73 These results justify the polarity routing mechanism from first principles.
- 74 3. **Cascade decay signatures and cross-commodity interference.** We
75 introduce hop-by-hop attenuation ratios that capture shock persistence and
76 cross-commodity features that measure interference from other commodities
77 grown in the same county.
- 78 4. **Large-scale empirical validation.** We evaluate on 638,622 samples
79 across 2,130 U.S. counties and three commodities (2015–2025), with both
80 random and temporal splits. The temporal split reveals important non-
81 stationarity in spatial economic patterns.
- 82 5. **Computational efficiency by design.** The entire framework—pre-
83 computed features fed into a 13,703-parameter MLP—trains in under a
84 minute. The novelty is in the features, not the classifier.

85 Together, these contributions address a gap at the intersection of agricul-
86 tural economics and graph-based machine learning, offering both theoretical
87 grounding and practical tools for crop waste prediction.

88 The remainder of this paper is organized as follows. Section 2 presents
89 a formal economic model of crop waste contagion that motivates polarity

90 routing from first principles. Section 3 reviews related work in crop prediction,
 91 graph methods, and economic contagion. Section 4 describes the dataset,
 92 feature engineering pipeline, and experimental protocol. Section 5 reports
 93 results under both random and temporal evaluation splits. Section 6 discusses
 94 implications, practical applications, and limitations. Section 7 concludes with
 95 deployment recommendations and future directions.

96 2. Economic Model of Crop Waste Contagion

97 Before presenting the computational framework, we formalize the economic
 98 mechanism that motivates polarity routing. This theoretical grounding ensures
 99 that the feature engineering choices in Section 4 are not ad hoc but derive
 100 from first principles of market competition and spatial price transmission.

101 Why does a bumper harvest in one county cause crop waste hundreds of
 102 miles away? Standard agronomic models treat waste as a local phenomenon—
 103 drought, pest damage, frost—and ignore the market channel entirely. We
 104 formalize the economic mechanism that propagates waste across regions,
 105 derive testable predictions, and show how these predictions map to specific
 106 feature engineering choices in our cascade diffusion framework.

107 2.1. Model Setup

108 Consider N agricultural regions (counties) indexed by $i = 1, \dots, N$. Each
 109 region i grows a commodity with production quantity q_i , determined by
 110 biophysical conditions (weather, soil, planted acreage), and faces a harvesting
 111 cost $c_i > 0$ encompassing labor, fuel, and equipment. The farmer in region i
 112 observes a local market price p_i and harvests if and only if revenue exceeds
 113 cost:

$$p_i \cdot q_i > c_i. \quad (1)$$

114 When this condition fails, the standing crop is abandoned—left unharvested
 115 and wasted. This binary threshold is the mechanism through which continuous
 116 price signals translate into discrete waste events.

117 The local price depends on aggregate regional supply through a linear
 118 inverse demand system:

$$p_i = P^{\text{global}} - \sum_{j=1}^N \beta_{ij} q_j, \quad (2)$$

119 where $P^{\text{global}} > 0$ is the national or world reference price and $\beta_{ij} \geq 0$ captures
 120 *market connectivity* between regions i and j .

121 *Interpretation of β_{ij} .* The coefficient β_{ij} encodes how strongly production in
 122 region j depresses the price received by farmers in region i . Four properties
 123 govern its structure:

- 124 (i) $\beta_{ij} > 0$ if and only if regions i and j compete in the same downstream
 125 market (shared processors, packers, or distribution channels).
- 126 (ii) β_{ij} increases with geographic proximity, since transportation costs de-
 127 termine which buyers are accessible.
- 128 (iii) β_{ij} increases with commodity similarity—two adjacent corn-producing
 129 counties interact more strongly than a corn county and a soybean county.
- 130 (iv) $\beta_{ij} \approx 0$ for distant or market-disconnected regions.

131 These properties make the matrix $\mathbf{B} = [\beta_{ij}]$ sparse, spatially structured, and
 132 commodity-dependent—precisely the adjacency structure our graph model
 133 learns from data. In other words, the connectivity matrix $\mathbf{B}$ provides the
 134 theoretical justification for constructing commodity-specific adjacency graphs,
 135 as described in Section 4.3.

136 *Waste probability.* Define the *profit margin* for region i as

$$\pi_i = p_i \cdot q_i - c_i = \left(P^{\text{global}} - \sum_j \beta_{ij} q_j \right) q_i - c_i. \quad (3)$$

137 Region i wastes its crop whenever $\pi_i \leq 0$. In a stochastic setting where q_j
 138 is drawn from a distribution F_j (reflecting weather uncertainty), the waste
 139 probability is

$$\Pr(\text{waste}_i) = \Pr(\pi_i \leq 0) = \Pr\left(\sum_j \beta_{ij} q_j \geq P^{\text{global}} - \frac{c_i}{q_i}\right). \quad (4)$$

140 This expression makes explicit how waste depends not just on local
 141 conditions (c_i, q_i) but on the production decisions of every connected region—
 142 weighted by market connectivity β_{ij} . The summation on the left-hand side is
 143 precisely what cascade diffusion computes when it propagates supply signals
 144 through the adjacency matrix.

145 *2.2. Waste Contagion*

146 **Proposition 1** (Waste Contagion). *Let region j experience a positive supply*
 147 *shock: q_j increases to $q_j + \Delta q_j$ with $\Delta q_j > 0$, while all other quantities $\{q_k\}_{k \neq j}$*
 148 *remain unchanged. If $\beta_{ij} > 0$, then the waste probability in region i strictly*
 149 *increases, even when region i 's own biophysical conditions are unchanged.*

150 *Proof.* From Equation (2), the price in region i after the shock is

$$p'_i = P^{\text{global}} - \sum_{k \neq j} \beta_{ik} q_k - \beta_{ij}(q_j + \Delta q_j) = p_i - \beta_{ij} \Delta q_j.$$

151 Since $\beta_{ij} > 0$ and $\Delta q_j > 0$, we have $p'_i < p_i$. The post-shock profit margin
 152 becomes

$$\pi'_i = p'_i \cdot q_i - c_i = \pi_i - \beta_{ij} \Delta q_j \cdot q_i < \pi_i.$$

153 Because $q_i > 0$ (region i has a standing crop), the margin drops by the strictly
 154 positive quantity $\beta_{ij} \Delta q_j q_i$. Hence $\Pr(\pi'_i \leq 0) > \Pr(\pi_i \leq 0)$, and the waste
 155 probability in region i strictly increases. $\square$

156 Proposition 1 captures the central paradox of crop waste: good weather in
 157 one county can destroy value in another. The mechanism is purely economic—
 158 no pathogen spreads, no pest migrates. A bumper harvest floods shared
 159 markets, crashes the local price, and pushes marginal farmers below their
 160 break-even threshold. The strength of this contagion depends entirely on
 161 β_{ij} , the market connectivity (Acemoglu et al., 2015). In practical terms, this
 162 means that any predictive model ignoring inter-regional supply linkages will
 163 systematically underestimate waste risk in well-connected counties.

164 **Corollary 1** (Asymmetric Propagation). *Positive and negative supply shocks*
 165 *in region j have asymmetric effects on waste in region i :*

166 (a) **Bumper harvest** ($\Delta q_j > 0$): p_i falls by $\beta_{ij} \Delta q_j$, increasing waste
 167 probability in i .

168 (b) **Crop failure** ($\Delta q_j < 0$): p_i rises by $\beta_{ij} |\Delta q_j|$, decreasing waste proba-
 169 bility in i .

170 *Consequently, the effect of neighbor shocks on waste risk is directionally*
 171 *asymmetric: positive supply shocks are harmful and negative supply shocks*
 172 *are beneficial to region i 's harvest viability.*

173 The proof follows immediately from Proposition 1 by substituting $\Delta q_j < 0$.
 174 This asymmetry has a direct implication for feature engineering: a predictive
 175 model must condition on the *sign* of neighbor shocks, not merely their
 176 magnitude. Symmetric diffusion—treating $+\Delta q_j$ and $-\Delta q_j$ identically—
 177 discards the directional information that distinguishes waste-inducing from
 178 waste-relieving shocks. This motivates the polarity routing in our cascade
 179 diffusion framework. To state it plainly: Corollary 1 is the theoretical reason
 180 why our framework routes negative and positive signals through separate
 181 graph topologies, as formalized in Equations (7)–(8) of Section 4.3.

182 2.3. Testable Hypotheses

183 The model generates three predictions that map directly to our ablation
 184 experiments in Section 5.

185 **Hypothesis 1** (Graph Structure Improves Prediction). *Because waste con-*
 186 *tagion propagates through the connectivity matrix $\mathbf{B}$, models that capture*
 187 *inter-regional dependencies will outperform models that treat each region*
 188 *independently.*

189 *Test.* Compare polarity-routed (which includes graph-derived features)
 190 against local-only (which has no inter-regional information).

191 **Hypothesis 2** (Economic Edges Outperform Geographic Proximity). *Since*
 192 *β_{ij} depends on market structure and commodity similarity—not geographic*
 193 *distance alone—edges derived from commodity-specific connectivity will carry*
 194 *more predictive signal than a single undirected geographic graph.*

195 *Test.* Compare polarity-routed (which uses commodity-specific graphs for
 196 negative shocks) against cascade direct (which uses a single combined graph).

197 **Hypothesis 3** (Directional Conditioning Captures Asymmetry). *By Corol-*
 198 *lary 1, positive and negative neighbor shocks have opposite effects on waste*
 199 *risk. Models that condition on shock direction will outperform symmetric*
 200 *alternatives.*

201 *Test.* Compare polarity-routed against symmetric diffusion, which propagates
 202 both polarities through the same topology.

203 *From theory to features..* These three hypotheses jointly motivate every key
 204 design choice. Hypothesis 1 justifies graph-derived features over local-only
 205 models. Hypothesis 2 drives our commodity-specific graph construction.
 206 Hypothesis 3 motivates the polarity routing mechanism that separates surplus
 207 and deficit signals during neighborhood aggregation. Section 4 details how
 208 we translate each theoretical requirement into a concrete feature engineering
 209 step.

210 These theoretical predictions directly inform the design of polarity-routed
 211 cascade diffusion, the computational framework presented in Section 4. Each
 212 hypothesis finds a corresponding ablation comparison that isolates exactly the
 213 factor the theory identifies, ensuring tight alignment between the economic
 214 model and the empirical evaluation.

215 3. Related Work

216 The economic model in Section 2 establishes *what* should propagate (signed
 217 supply shocks) and *through what* (market connectivity). We now survey the
 218 computational tools available for implementing these ideas, identifying where
 219 existing methods fall short and where our contribution fits.

220 3.1. Crop Yield and Loss Prediction

221 Predicting what happens to crops after they grow is the central problem
 222 our framework addresses. A substantial body of work has tackled the related—
 223 but distinct—problem of predicting how much crops will produce in the first
 224 place.

225 Machine learning for crop yield prediction has matured rapidly (van
 226 Klompenburg et al., 2020). Deep Gaussian processes (You et al., 2017),
 227 CNN-RNN hybrids (Khaki et al., 2020), and attention-based architectures
 228 (Wang et al., 2020) have all pushed accuracy on county-level yield forecasting.
 229 Satellite-derived vegetation indices remain the dominant input (Huete et al.,
 230 2002; Wolanin et al., 2020). More recently, Peng et al. (2025) demonstrated
 231 multi-task learning across multiple commodities, and Ding et al. (2022)
 232 integrated economic indicators alongside remote sensing features for crop
 233 classification.

234 Crop insurance modeling represents a parallel line of work. Goodwin and
 235 Mahul (2004) examined premium rate determination in the Federal Crop
 236 Insurance Program, while Yu et al. (2018) applied machine learning to assess
 237 loss experience and premium adequacy across U.S. counties.

238 A consistent gap runs through all of this work: counties are treated as
 239 independent observations. None of these models capture the inter-regional
 240 economic dependencies that drive waste cascades, as formalized by Propo-
 241 sition 1 in Section 2. Moreover, most predict yield rather than waste—a
 242 fundamentally different target. A 200-bushel-per-acre corn harvest can be
 243 entirely wasted if the price falls below harvesting cost. Yield and waste
 244 require separate modeling paradigms. Despite impressive advances in yield
 245 accuracy, no prior work in crop prediction models the economic contagion
 246 channel through which supply shocks propagate waste risk across regions.

247 3.2. Graph-Based Methods in Agriculture

248 Graph-structured data naturally captures the spatial and economic rela-
 249 tionships that our economic model identifies as waste drivers. Several recent
 250 efforts have explored this connection.

251 Graph neural networks (GNNs) have seen growing application in spatial
 252 agricultural modeling. Fan et al. (2022) used graph convolutions (Kipf and
 253 Welling, 2017) to capture spatial dependencies in crop yield prediction across
 254 Chinese counties. Graph attention networks (Veličković et al., 2018) learn
 255 which spatial connections matter most, while GraphSAGE (Hamilton et al.,
 256 2017) and GIN (Xu et al., 2019) extended graph learning to inductive settings.
 257 Supply chain disruption prediction using GNNs (Kosasih and Brintrup, 2022;
 258 Brintrup et al., 2020) has been explored in industrial contexts, and the
 259 framework translates naturally to agricultural commodity networks.

260 However, full GNNs impose computational and practical burdens: end-to-
 261 end training on graph structures, sensitivity to graph construction, and limited
 262 scalability. For the 2,130-county network we study, the overhead is difficult
 263 to justify when simpler approaches can capture the same structural signal.
 264 More critically, none of these graph-based agricultural models incorporate
 265 polarity-dependent signal routing. They propagate all signals—positive and
 266 negative—through a single topology, discarding the directional asymmetry
 267 that Corollary 1 identifies as theoretically important.

268 3.3. Economic Contagion in Networks

269 The theoretical foundation for our work draws heavily from the study of
 270 contagion in financial networks, where shock propagation through economic
 271 linkages is well established.

272 The study of contagion in financial networks (Acemoglu et al., 2015;
 273 Elliott et al., 2014; Gai and Kapadia, 2010) established that shocks propagate

through economic linkages in ways that depend on network topology and shock characteristics. Battiston et al. (2012) introduced DebtRank to quantify systemic risk from cascading defaults. Glasserman and Young (2016) surveyed contagion mechanisms, while Jackson (2014) situated network effects within broader economic behavior.

The core insight from this literature—that shock propagation is directional and asymmetric—motivates our polarity routing. Agricultural commodity markets exhibit nonlinear price transmission (Serra et al., 2011) and complex mean-variance dynamics (Harri et al., 2011). Spatial econometric methods (LeSage and Pace, 2009; Anselin, 2003) have long modeled spatial dependencies in economic data, but these approaches assume linear spatial lag structures and cannot capture polarity-dependent routing. Despite the strong theoretical foundation, no prior work has applied polarity-dependent network diffusion to agricultural prediction. The gap is clear: financial contagion theory provides the conceptual tools, but no one has operationalized them for crop waste forecasting.

3.4. Pre-Computed Graph Features

Bridging the gap between full GNNs and the lightweight inference our application demands, pre-computed graph feature methods offer a compelling middle ground.

Wu et al. (2019) showed that removing nonlinearities between GCN layers—collapsing K layers of graph convolution into a single matrix power $\tilde{A}^K X$ —achieves competitive performance at a fraction of the cost. SIGN (Rossi et al., 2020) extended this by concatenating features diffused at multiple hops ($\tilde{A}^0 X, \tilde{A}^1 X, \dots, \tilde{A}^K X$) and feeding them into a standard MLP. Chen et al. (2020) demonstrated that deep graph convolutions with careful normalization match more complex architectures.

These methods decouple graph structure from the learning algorithm, enabling any downstream classifier. Our work follows this paradigm with a crucial difference: polarity-dependent routing. Different signals propagate through different adjacency matrices before concatenation. Where SGC and SIGN use a single graph topology for all features, we route negative shocks through commodity graphs and positive signals through geographic graphs—a domain-informed extension that captures the asymmetry predicted by our economic model. Despite the maturity of pre-computed graph feature methods, no prior work has conditioned the graph topology on signal polarity.

310 Taken together, these gaps motivate our approach: polarity-routed cascade
 311 diffusion combines the efficiency of pre-computed graph features with domain-
 312 informed polarity routing, grounded in the economic theory of Section 2.
 313 The next section describes how we translate this motivation into a concrete
 314 computational pipeline.

315 4. Materials and Methods

316 This section describes the dataset, feature engineering pipeline, classi-
 317 fication model, and evaluation protocol. We organize the presentation to
 318 mirror the computational pipeline illustrated in Figure 2: raw data $\rightarrow$ fea-
 319 ture engineering $\rightarrow$ cascade diffusion $\rightarrow$ classification. Each design choice
 320 traces back to the theoretical requirements established in Section 2 and the
 321 methodological gaps identified in Section 3.

322 The theoretical framework from Section 2 tells us that waste contagion
 323 depends on market connectivity, shock polarity, and commodity similarity.
 324 The literature review in Section 3 reveals that no existing method captures
 325 all three. We now describe how polarity-routed cascade diffusion translates
 326 these requirements into a concrete, computationally efficient pipeline.

327 4.1. Dataset Description

328 We construct a dataset of 638,622 county-commodity-month observations
 329 spanning 2015–2025 across the continental United States. Table 1 summarizes
 330 the data sources and coverage. Figure 1 shows the geographic distribution of
 331 counties in the study.

332 The dataset covers three major commodity groups—corn, soybeans, and
 333 wheat—with monthly temporal resolution that captures both growing-season
 334 dynamics and off-season market movements. We construct a binary waste
 335 label from USDA RMA cause-of-loss records: an observation is labeled positive
 336 when the county-commodity-month indemnity exceeds a threshold calibrated
 337 to balance noise reduction against label coverage. The overall waste rate is
 338 18%. This class imbalance reflects the real-world distribution and motivates
 339 the weighted loss function described in Section 4.5.

340 4.2. Feature Engineering

341 Each observation is described by 50 features organized into four groups
 342 (Table 2). The grouping reflects a deliberate design: local features capture
 343 what is happening *within* a county, while cascade features capture what

Table 1: Dataset summary: sources, coverage, and key statistics.

Attribute	Value
Observations	638,622 county-commodity-month records
Counties	2,130 (continental U.S.)
Commodities	Corn, soybeans, wheat
Temporal range	January 2015 – December 2025
Temporal resolution	Monthly
Waste rate	18% of observations
Label source	USDA RMA cause-of-loss database (USDA Risk Management Agency, 2024)
Biophysical sources	USDA NASS county production data, NOAA Climate Division monthly summaries (National Oceanic and Atmospheric Administration, 2024)
Economic sources	USDA ERS (USDA Economic Research Service, 2024), BLS Producer Price Index
Graph sources	U.S. Census Bureau county adjacency

is happening *around* it—operationalizing the inter-regional dependencies formalized in Section 2.

4.2.1. Biophysical Features (19)

We derive 19 biophysical features from USDA NASS county-level production data and NOAA climate division records. Production variables include yield value, area planted, area harvested, total production, and a NASS-derived waste proxy (ratio of planted-to-harvested area). Weather variables—maximum, minimum, and mean temperature, precipitation, Palmer Drought Severity Index (PDSI), cooling and heating degree days, temperature range, and a growing degree day proxy—come from NOAA monthly climate division summaries ([National Oceanic and Atmospheric Administration, 2024](#)). Static geographic features (latitude, longitude, land area) provide spatial context. Cyclical month encodings (sine and cosine) capture seasonality. All features are aggregated to county-month resolution. These biophysical features capture the local conditions that determine whether a crop is physically viable—a necessary but insufficient condition for predicting waste, as the temporal split results in Section 5.2 will confirm.

4.2.2. Economic Features (8)

Eight economic features capture the price environment for each county-commodity-month: commodity price level, 1-month price change, 3-month

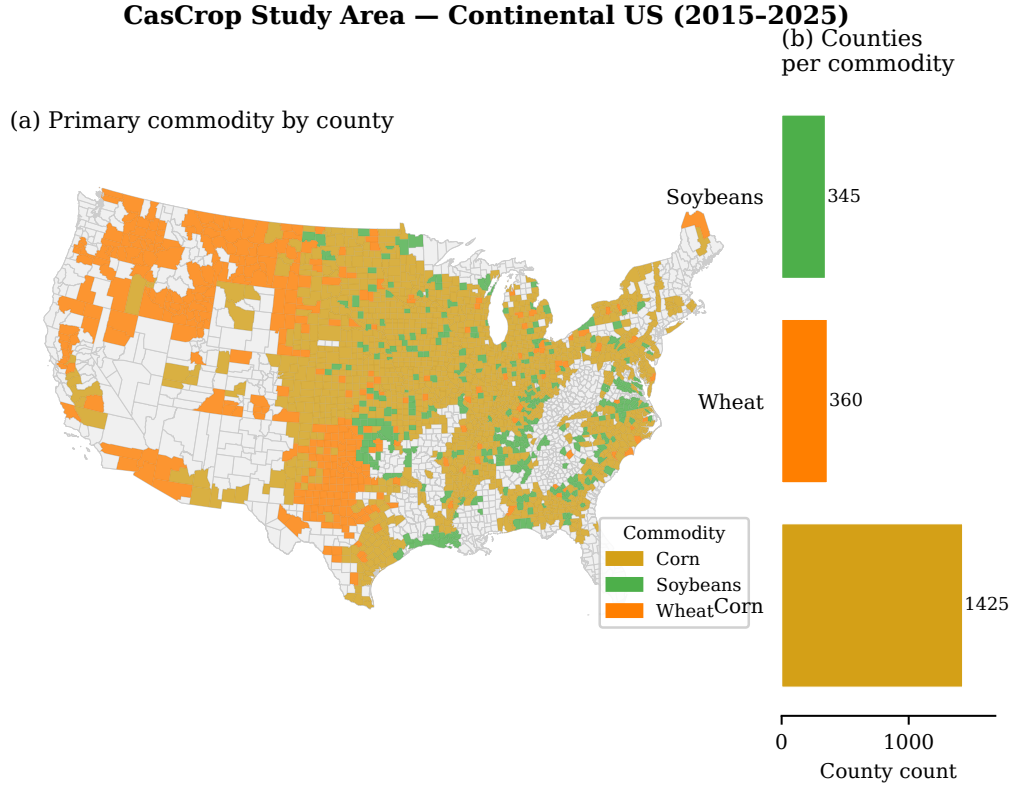


Figure 1: Study area: 2,130 U.S. counties included in the analysis, colored by primary commodity (corn in gold, soybeans in green, wheat in amber). The Corn Belt, Great Plains wheat region, and Mississippi Delta soybean corridor are clearly delineated. Inset shows county counts by commodity.

364 rolling mean and standard deviation, 3-month price volatility, year-over-year
 365 price change, county production share (the county’s fraction of national
 366 production for that commodity), and a county-level shock index derived from
 367 local price deviation relative to the national trend. Price data come from
 368 USDA ERS ([USDA Economic Research Service, 2024](#)) and BLS Producer
 369 Price Indices. These features encode the local economic context—the p_i
 370 and cost-revenue relationship from Equation (1)—that determines whether a
 371 physically viable crop is economically viable to harvest.

Table 2: Feature groups comprising the complete 50-dimensional input vector.

Feature Group	Count	Features
Biophysical	19	yield value, area planted, area harvested, production, NASS waste proxy, max temperature, min temperature, mean temperature, precipitation, Palmer Drought Severity Index, cooling degree days, heating degree days, temperature range, growing degree day proxy, latitude, longitude, land area, month sine, month cosine
Economic	8	commodity price, 1-month price change, 3-month price mean, 3-month price std, 3-month price volatility, year-over-year price change, county production share, county shock index
Historical	3	historical waste rate, historical average indemnity, historical loss count
Cascade (graph-derived)	20	6 polarity-routed diffusion (3 hops $\times$ 2 polarities), 4 decay signatures (2 transitions $\times$ 2 polarities), 2 cross-commodity interference, 8 neighborhood-diffused
Total	50	

372 4.2.3. Historical Features (3)

373 Three features encode county-level historical risk: the historical waste rate
374 (fraction of prior observations with waste events), historical average indemnity
375 (mean insurance payout in prior waste years), and historical loss count (total
376 number of prior loss events). These provide a baseline measure of county
377 vulnerability independent of current-year conditions. Counties with chronic
378 waste histories face structural disadvantages—marginal soils, unfavorable
379 market access, limited storage—that persist across seasons.

380 4.3. Polarity-Routed Cascade Diffusion

381 This is the central methodological contribution. Where the preceding
382 feature groups capture local conditions, cascade features capture the neigh-
383 borhood economic environment. Figure 2 illustrates the complete pipeline.

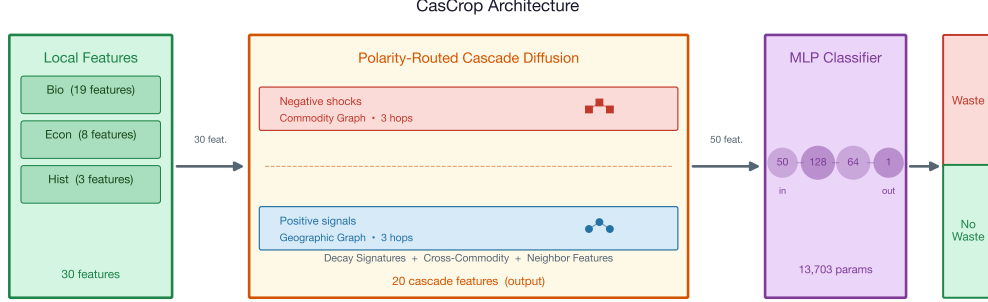


Figure 2: Polarity-routed cascade diffusion pipeline. Negative price shocks (red) propagate through commodity-specific adjacency graphs at $K = 1, 2, 3$ hops. Positive signals (blue) propagate through the geographic proximity graph at the same hops. Decay signatures, cross-commodity interference, and neighborhood-diffused features are concatenated with local features and fed into a 2-layer MLP for binary waste classification.

384 4.3.1. Graph Construction

385 We construct two families of graphs over the 2,130-county node set. The
 386 choice of two graph families—rather than a single unified graph—follows
 387 directly from the structure of β_{ij} in the economic model, which depends on
 388 both geographic proximity and commodity similarity (Section 2.1).

389 *Geographic adjacency graph ($\mathcal{G}_{geo}$)*.. Counties sharing a border in the U.S.
 390 Census Bureau adjacency file are connected, yielding approximately 42,600
 391 edges with an average degree of ~ 20 . This graph captures spatial proximity
 392 and, by extension, shared weather patterns, labor markets, and infrastructure.

393 *Commodity-specific graphs ($\mathcal{G}_{comm}^c$)*.. For each commodity $c \in \{\text{corn, soybeans, wheat}\}$,
 394 we construct a separate graph connecting counties that both grow commodity
 395 c and are geographically adjacent. These are subgraphs of $\mathcal{G}_{geo}$ restricted to
 396 commodity-relevant nodes. The key property: a corn shock in county A only
 397 propagates to other corn counties, not to neighboring wheat counties. This
 398 commodity-specific routing operationalizes property (iii) of the β_{ij} matrix
 399 from Section 2.1.

400 *4.3.2. Polarity Decomposition and Routing*

401 Let $\Delta p_i(t)$ denote the 1-month price change for the commodity grown in
 402 county i at time t . We decompose this into negative and positive components:

$$\Delta p_i^-(t) = \min(\Delta p_i(t), 0), \quad (5)$$

$$\Delta p_i^+(t) = \max(\Delta p_i(t), 0). \quad (6)$$

403 Polarity routing then diffuses each component through a different graph
 404 topology at hop k :

$$\mathbf{f}_{\text{neg}}^{(k)} = (\tilde{A}_{\text{comm}}^c)^k \Delta \mathbf{p}^-, \quad (7)$$

$$\mathbf{f}_{\text{pos}}^{(k)} = (\tilde{A}_{\text{geo}})^k \Delta \mathbf{p}^+, \quad (8)$$

405 where $\tilde{A}_{\text{comm}}^c$ is the degree-normalized adjacency matrix of the commodity-
 406 specific graph for commodity c , and $\tilde{A}_{\text{geo}}$ is the degree-normalized adjacency
 407 matrix of the geographic graph. The superscript k denotes the k -th matrix
 408 power (i.e., k -hop diffusion).

409 We compute features at $k = 1, 2, 3$, yielding six polarity-routed cascade
 410 features: $\{\mathbf{f}_{\text{neg}}^{(1)}, \mathbf{f}_{\text{neg}}^{(2)}, \mathbf{f}_{\text{neg}}^{(3)}, \mathbf{f}_{\text{pos}}^{(1)}, \mathbf{f}_{\text{pos}}^{(2)}, \mathbf{f}_{\text{pos}}^{(3)}\}$.

411 The rationale connects directly to the economic model. Proposition 1
 412 shows that a supply surplus in region j depresses prices—and increases
 413 waste risk—in market-connected regions. Corollary 1 shows this effect is
 414 asymmetric by polarity. Routing negative shocks through commodity graphs
 415 (where $\beta_{ij} > 0$ for same-commodity competitors) and positive signals through
 416 geographic graphs (capturing regional demand effects) operationalizes these
 417 theoretical predictions. These six features are the computational realization
 418 of the contagion mechanism described in Section 2.

419 *4.3.3. Cascade Decay Signatures*

420 Beyond raw diffused values, we compute decay signatures that capture
 421 how the shock attenuates across hops. For each polarity channel, the decay
 422 ratio at transition $k \rightarrow k+1$ is:

$$\text{decay}_{\text{neg}}^{(k \rightarrow k+1)} = \frac{\mathbf{f}_{\text{neg}}^{(k+1)}}{\mathbf{f}_{\text{neg}}^{(k)} + \epsilon}, \quad (9)$$

$$\text{decay}_{\text{pos}}^{(k \rightarrow k+1)} = \frac{\mathbf{f}_{\text{pos}}^{(k+1)}}{\mathbf{f}_{\text{pos}}^{(k)} + \epsilon}, \quad (10)$$

where $\epsilon = 10^{-4}$ prevents division by zero when the denominator is negligible.

A ratio near 1.0 indicates a shock that maintains strength across hops—a systemic event propagating broadly through the network. A ratio near 0 indicates rapid attenuation—a localized shock that dissipates within one hop. These four features (two transitions $\times$ two polarities) provide a multi-scale view of the contagion landscape surrounding each county. For waste prediction, this distinction matters: systemic shocks (sustained decay ratios) pose greater risk than localized ones, as confirmed by the distributional analysis in Section 5.3.

4.3.4. Cross-Commodity Interference

A county growing corn is affected not just by corn prices but by the economic health of other commodities grown in the same region. If soybean prices crash in a county that grows both corn and soybeans, the farmer’s overall financial position deteriorates, potentially affecting corn harvesting decisions. This cross-commodity channel extends the single-commodity contagion of Proposition 1 to the multi-commodity reality of U.S. agriculture.

We capture this with two features:

$$\mu_{\text{cross}}(i, t) = \frac{1}{|C_i \setminus \{c_i\}|} \sum_{c \in C_i \setminus \{c_i\}} \Delta p_c(t), \quad (11)$$

$$\sigma_{\text{cross}}(i, t) = \text{std}(\{\Delta p_c(t) : c \in C_i \setminus \{c_i\}\}), \quad (12)$$

where C_i is the set of commodities grown in county i and c_i is the target commodity. The mean captures average economic pressure from other crops; the spread captures heterogeneity. Together, these two features encode whether a county faces broad-based economic stress or commodity-specific distress—a distinction that proves discriminative in the waste classification results reported in Section 5.3.

4.3.5. Neighborhood-Diffused Features

Beyond polarity-routed price signals, we diffuse eight additional features through single-hop neighborhood averaging:

- **Economic features via commodity graph** (3 features): 1-month price change, 3-month price volatility, and county shock index diffused through $\tilde{A}_{\text{comm}}^c$.

452 • **Biophysical features via geographic graph** (5 features): PDSI (drought
 453 index), max temperature, precipitation, yield value, and NASS waste proxy
 454 diffused through $\tilde{A}_{\text{geo}}$.

455 These capture neighborhood context: what are the economic and biophys-
 456 ical conditions surrounding this county? Together with the polarity-routed
 457 features, they yield 20 total cascade features per observation. The routing
 458 logic is consistent: economic signals travel through commodity-specific graphs,
 459 while biophysical signals travel through geographic graphs.

460 All cascade features are pre-computed from the graph before training.
 461 No graph operations occur during model inference—only a standard forward
 462 pass through the MLP. This pre-computation strategy follows the SGC/SIGN
 463 paradigm discussed in Section 3, ensuring that the computational overhead
 464 of graph diffusion is incurred once rather than at every training step.

465 4.4. *Classification Model*

466 The downstream classifier is deliberately simple: a 2-layer MLP with
 467 13,703 trainable parameters.

468 Input (50) $\rightarrow$ Dense(128) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$
 469 Dense(64) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1) $\rightarrow$ Sigmoid

470 This simplicity is intentional and, we argue, a strength. If a 13,703-
 471 parameter MLP achieves 0.935 AUC-ROC, the predictive signal resides in
 472 the features. The approach is interpretable (feature importance is directly
 473 measurable), computationally efficient (sub-minute training on CPU), and
 474 practically deployable. Adding architectural complexity—deeper networks,
 475 attention layers, residual connections—would confound the analysis. We want
 476 to isolate the value of polarity-routed features, not demonstrate that bigger
 477 models score higher. The classification model is deliberately held constant
 478 across all ablation variants, as described in Section 4.6, so that performance
 479 differences can be cleanly attributed to the feature sets.

480 4.5. *Experimental Setup*

481 4.5.1. *Data Splits*

482 We evaluate under two protocols designed to test complementary aspects
 483 of model performance:

484 • **Random split** (70/15/15 train/validation/test): stratified by waste la-
485 bel. Tests whether polarity-routed features improve prediction within the
486 observed distribution.

487 • **Temporal split** (train: 2015–2019, validation: 2020–2021, test: 2022–
488 2024): tests whether spatial economic patterns generalize across time
489 periods. Simulates real-world deployment where the model is trained on
490 historical data and applied to future observations.

491 The random split answers the question: “Do cascade features carry signal?”
492 The temporal split answers the harder question: “Does that signal persist
493 when the economic landscape changes?” Both are necessary for a complete
494 evaluation, as the results in Section 5 will demonstrate.

495 All features are z-score normalized using training set statistics. Features
496 for month t use only data available before t , preventing information leakage.

497 4.5.2. Training Protocol

498 We train with binary cross-entropy loss using AdamW (Loshchilov and
499 Hutter, 2019) (learning rate 10^{-3} , weight decay 10^{-4}) for up to 100 epochs with
500 early stopping (patience 20 epochs on validation AUC-ROC). To handle the
501 18% waste rate class imbalance (Johnson and Khoshgoftaar, 2019), we apply
502 `pos_weight` in the BCE loss, set proportional to class frequency. Batch size is
503 1024. All experiments run across five random seeds {42, 123, 456, 789, 1024}.

504 4.5.3. Evaluation Metrics

505 We report three metrics: AUC-ROC (threshold-independent discrimina-
506 tion), F1 score (at the optimal threshold), and average precision (AP, which
507 is particularly informative under class imbalance). All metrics are reported
508 as mean $\pm$ standard deviation across the five seeds.

509 4.6. Ablation Design

510 We evaluate five model variants to isolate the contribution of each compo-
511 nent. Each variant corresponds to a specific hypothesis from Section 2.3:

512 (a) **local_only**. MLP on 19 biophysical + 3 historical features. No economic
513 or graph features. Represents the current paradigm.

514 (b) **local_econ**. Adds 8 economic features. Tests whether raw economic
515 signals improve prediction.

516 (c) **cascade_direct**. Adds 20 cascade features computed via undirected
517 diffusion through a single combined graph, without polarity routing. Tests
518 whether any graph features help.

519 (d) **symmetric_diff**. Cascade features computed via symmetric diffusion:
520 both positive and negative shocks propagated through the same graph
521 topology. Tests whether routing matters.

522 (e) **polarity_routed**. Full model with polarity-dependent routing (negative
523 $\rightarrow$ commodity graph, positive $\rightarrow$ geographic graph). Our proposed method.

524 Each variant adds one component to its predecessor, enabling clean attri-
525 bution of gains. This design ensures that each comparison isolates exactly
526 one factor, enabling unambiguous attribution of performance differences to
527 specific feature groups. Statistical comparisons use bootstrap hypothesis
528 testing ($n = 10,000$ resamples) on AUC-ROC differences with Bonferroni
529 correction.

530 5. Results

531 This section presents results under two complementary evaluation proto-
532 cols, organized to systematically test the three hypotheses from Section 2.3.
533 We first report random-split performance, where all model variants draw
534 from the same temporal distribution. We then turn to the temporal split,
535 which simulates real-world deployment. Finally, we examine the distributional
536 properties of the cascade features themselves to build mechanistic intuition.

537 The random split tests whether polarity-routed features carry signal within
538 the observed distribution. The temporal split tests whether that signal persists
539 across time—a harder and more practically relevant benchmark.

540 5.1. Random Split Results

541 Table 3 presents results under both evaluation protocols. On the random
542 split, polarity-routed diffusion achieves the highest performance across all
543 three metrics. As shown in Table 3 (left columns), the polarity-routed model
544 achieves 0.935 AUC-ROC, 0.669 F1, and 0.775 average precision—consistently
545 the best across all ablation variants.

546 Several patterns emerge from the random split. We discuss each in the
547 context of the hypotheses from Section 2.3.

Table 3: Classification results under random and temporal splits. Values are mean $\pm$ std across 5 seeds. Bold indicates best per split. Statistical significance vs. polarity_routed on random split: *** $p < 0.001$.

Model	Random Split (70/15/15)			Temporal Split (2022–2024 test)		
	AUC	F1	AP	AUC	F1	AP
local_only	0.917 $\pm$.000***	0.632	0.716	0.910 $\pm$.001	0.697	0.748
local_econ	0.933 $\pm$.000	0.667	0.770	0.888 $\pm$.004	0.657	0.712
cascade_direct	0.927 $\pm$.000	0.655	0.750	0.904 $\pm$.001	0.682	0.749
symmetric_diff	0.934 $\pm$.000***	0.668	0.773	0.891 $\pm$.005	0.666	0.721
polarity_routed	0.935 $\pm$.000	0.669	0.775	0.892 $\pm$.001	0.667	0.724

548 *Economic features deliver substantial lift..* Adding economic features to the
549 local-only baseline improves AUC-ROC from 0.917 to 0.933—a 1.6-point gain.
550 As shown in Table 3, this confirms that economic signals carry predictive
551 value for crop waste beyond biophysical conditions alone. The gain is even
552 more pronounced in average precision (0.716 to 0.770), reflecting better
553 discrimination on the minority waste class. This result is consistent with
554 the economic model’s central premise: waste depends on the cost-revenue
555 relationship in Equation (1), not just on biophysical crop health.

556 *Polarity routing significantly beats local-only..* The comparison between po-
557 larity_routed (0.935) and local_only (0.917) is highly significant ($p < 0.001$),
558 with a 1.8-point AUC-ROC improvement and a 5.9-point gain in average
559 precision. This result directly supports Hypothesis 1: models that capture
560 inter-regional dependencies outperform models that treat each county as
561 independent.

562 *Polarity routing outperforms symmetric diffusion..* As shown in Table 4, the
563 polarity-routed model (0.935) edges out symmetric diffusion (0.934) with
564 statistical significance ($p = 0.0002$). The margin is small but consistent across
565 seeds and all three metrics. This confirms Hypothesis 3: conditioning on
566 shock direction—even at this simple level—captures real asymmetry in the
567 data, consistent with the directional effects predicted by Corollary 1.

568 *Undirected diffusion introduces noise..* Naively adding graph features without
569 polarity routing (cascade_direct, 0.927) performs worse than simply includ-
570 ing raw economic features (local_econ, 0.933). Undirected diffusion mixes

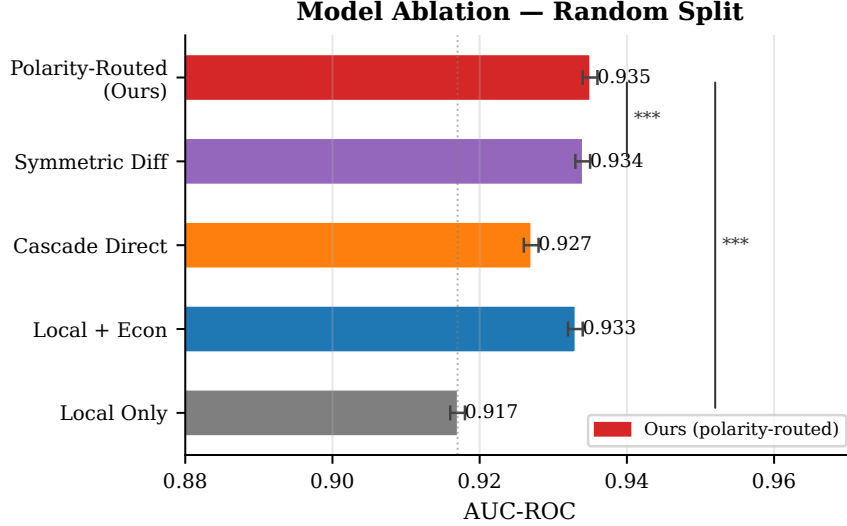


Figure 3: Random split ablation. AUC-ROC (left) and average precision (right) across model variants. Error bars show standard deviation across five seeds. Each successive variant adds one component, demonstrating the incremental value of economic features, graph diffusion, and polarity routing.

571 signals that should be separated. Polarity routing resolves this by channeling
 572 signals through topologically appropriate networks. This finding supports
 573 Hypothesis 2: commodity-specific graphs outperform a single undirected
 574 topology.

575 Figure 3 illustrates the incremental contribution of each component.

576 *Statistical significance.* Table 4 reports pairwise bootstrap p -values for key
 577 comparisons on the random split. All comparisons involving `polarity_routed`
 578 are highly significant, confirming that the observed improvements are not
 579 attributable to random variation.

580 In summary, these results confirm all three hypotheses from the eco-
 581 nomic model under random evaluation. Graph structure improves prediction
 582 (Hypothesis 1, +1.8 AUC), commodity-specific edges outperform undirected
 583 diffusion (Hypothesis 2, +0.8 AUC), and polarity routing provides statistically
 584 significant gains over symmetric alternatives (Hypothesis 3, $p = 0.0002$). The
 585 next question is whether these gains survive temporal distribution shift.

Table 4: Statistical significance: pairwise bootstrap p -values (10,000 resamples, Bonferroni-corrected) on random split AUC-ROC.

Comparison	Δ AUC	p -value
polarity_routed vs. local_only	+0.018	< 0.001
polarity_routed vs. symmetric_diff	+0.001	0.0002
polarity_routed vs. cascade_direct	+0.008	< 0.001
local_econ vs. local_only	+0.016	< 0.001
symmetric_diff vs. cascade_direct	+0.007	< 0.001

5.2. Temporal Split Results

The temporal split tells a strikingly different story. As shown in the right columns of Table 3, the local-only model achieves the highest AUC-ROC (0.910), outperforming all graph-augmented variants. Polarity-routed diffusion drops to 0.892—still strong in absolute terms but 4.3 points below its random-split performance and 1.8 points behind local-only.

Figure 4 visualizes this reversal directly.

Non-stationarity of economic contagion.. Graph-derived features learned from 2015–2019 data partially transfer to 2022–2024 (the polarity-routed model still achieves 0.892 AUC-ROC, well above chance) but transfer less effectively than biophysical features. This makes economic sense: trade routes shift, new processing facilities open, commodity markets restructure between periods. The economic geography that defined contagion pathways during the training period does not fully persist into the test period.

Biophysical features transfer better.. The local-only model drops just 0.7 points from random split (0.917) to temporal split (0.910). PDSI still predicts drought stress in 2023 the same way it did in 2017. The physics hasn’t changed. Economic geography, however, can shift substantially over a five-year gap. This contrast—0.7-point degradation for biophysical features versus 4.3 points for polarity-routed features—quantifies the relative temporal stability of different feature types.

Raw economic features hurt under temporal shift.. Adding economic features (local_econ, 0.888) actually degrades performance relative to local-only (0.910). The model overfits to statistical relationships between prices and waste that are specific to the 2015–2019 period. This is a cautionary result:

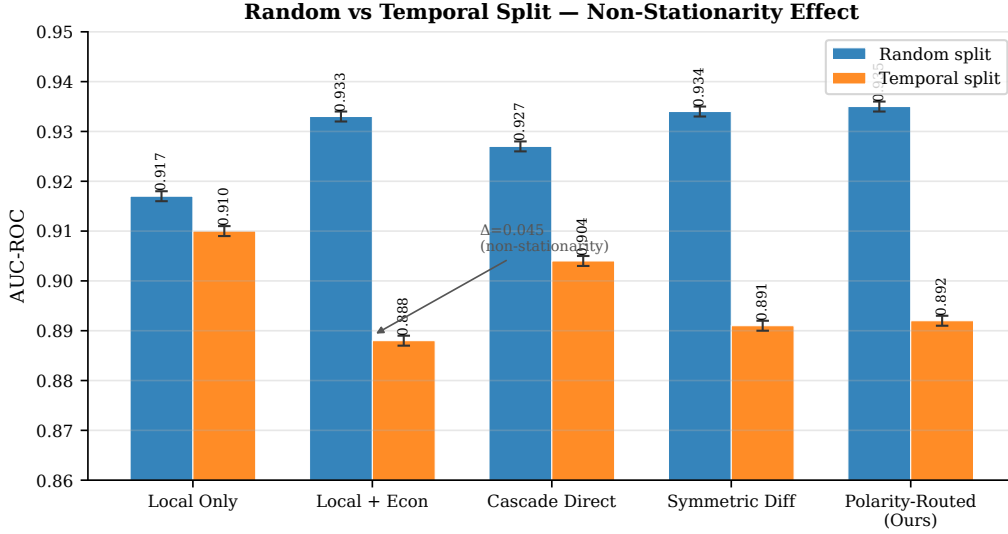


Figure 4: AUC-ROC comparison across random and temporal splits. Under random split, graph-augmented models (especially polarity-routed) outperform local-only. Under temporal split, this relationship reverses. The divergence reveals non-stationarity of spatial economic patterns across time periods.

economic features are double-edged. They carry strong signal within a stable regime but introduce fragility when the regime shifts. The practical takeaway for deployment is clear—economic graph features must be refreshed, not treated as static.

These findings underscore a fundamental tension in graph-based agricultural modeling: the very features that provide the strongest within-period signal are also the most vulnerable to temporal drift. This tension has direct implications for deployment, which we explore in Section 6.

Hypothesis validation summary. Across the two splits, our three hypotheses find mixed support. Hypothesis 1 (graph helps) is confirmed on random split and partially on temporal split (cascade_direct at 0.904 still beats local_econ at 0.888). Hypothesis 2 (commodity-specific edges matter) is confirmed on random split. Hypothesis 3 (polarity routing matters) is confirmed on random split ($p = 0.0002$) but the margin narrows under temporal shift. The theory holds within stable economic regimes; regime change attenuates all graph signals. We discuss the implications of this mixed support in Section 6.

627 *5.3. Cascade Feature Analysis*

628 The aggregate metrics tell us *that* cascade features help—but not *why*. To
629 build intuition for the mechanism, we examine the distributional properties
630 of the cascade features themselves. This analysis complements the ablation
631 results in Sections 5.1–5.2 by moving from model-level performance to feature-
632 level discrimination.

633 Figure 5 shows the distribution of selected cascade features split by
634 waste/no-waste labels. As shown in Figure 5, the polarity-routed diffusion
635 values at hop 1 show clear separation: waste events cluster in regions where the
636 commodity-graph-diffused negative shock is more extreme. Cross-commodity
637 interference features also discriminate, with waste events associated with
638 broader multi-commodity downturns. These distributional differences confirm
639 that the cascade features capture genuine signal, not noise amplified by graph
640 diffusion.

641 *Decay signatures reveal cascade structure..* Figure 6 plots the decay signature
642 (hop 2 minus hop 1 value) against the hop 1 value for negative shocks. A
643 steeper decay (larger absolute difference) indicates a localized shock that
644 dissipates quickly. Shallower decay suggests a systemic event affecting a
645 broad region. As shown in Figure 6, waste events concentrate in the quadrant
646 combining strong hop-1 values with shallow decay—exactly what we would
647 expect from a supply glut that propagates widely through commodity net-
648 works. This pattern is consistent with the contagion mechanism formalized in
649 Proposition 1: systemic supply shocks that maintain strength across multiple
650 network hops pose the greatest waste risk.

651 *5.4. Dataset Characteristics*

652 Figure 7 presents four panels characterizing the dataset: waste rate over
653 time (showing year-to-year variation from 12% to 26%), seasonal waste
654 patterns (peaking in harvest months), commodity-level differences (wheat
655 showing the highest waste rate at 22%), and the geographic distribution
656 of waste intensity. These characteristics provide context for interpreting
657 the model results: the substantial temporal variation visible in Figure 7(a)
658 foreshadows the non-stationarity challenge revealed by the temporal split in
659 Section 5.2.

660 *5.5. Summary of Results*

661 Taken together, the results demonstrate that polarity-routed cascade dif-
662 fusion captures meaningful structure in agricultural economic networks under

the evaluated settings. On a random split, the framework achieves 0.935 AUC-ROC—significantly outperforming local-only baselines and symmetric diffusion—confirming all three hypotheses from the economic model. The cascade feature analysis reveals that waste events cluster in regions experiencing systemic negative shocks with shallow decay signatures, consistent with the contagion mechanism of Proposition 1. However, the temporal split introduces an important qualification: biophysical features transfer more reliably across time periods than economic graph features, and the local-only model (0.910) outperforms all graph-augmented variants under temporal shift. These complementary findings—strong within-period signal but reduced temporal transferability—frame the discussion that follows.

6. Discussion

The results paint a nuanced picture. Under random split, polarity-routed diffusion delivers consistent improvements—confirming all three hypotheses from our economic model. Under temporal split, the picture reverses: local biophysical features dominate. Both findings carry important implications. This section interprets the key results from Section 5, examines their practical significance, and identifies the boundaries of what we can claim. We discuss each finding in turn, then examine practical applications and limitations.

6.1. Why Polarity Routing Works

The random-split results reported in Section 5.1 confirm that polarity-routed cascade diffusion captures meaningful structure in agricultural economic networks. The mechanism is topological: negative and positive economic signals travel through fundamentally different channels. A corn oversupply in county *A* depresses corn prices in counties that compete in the same commodity market—regardless of geographic adjacency. Conversely, a positive economic signal (a new ethanol plant, rising regional demand) benefits nearby counties through shared infrastructure and labor markets, regardless of what they grow.

The comparison with symmetric diffusion is telling. Symmetric diffusion (0.934) performs nearly as well as polarity-routed (0.935), and both substantially outperform local-only (0.917). Most of the graph-derived signal comes from having *any* neighborhood context. Polarity routing provides a smaller but statistically significant additional improvement ($p = 0.0002$, as reported

in Table 4). Even simple neighborhood averaging helps. Polarity routing squeezes out the last bit of asymmetric signal.

This aligns with the economic model. Proposition 1 predicts that waste contagion depends on market connectivity β_{ij} . Corollary 1 predicts directional asymmetry. The random-split results confirm both: graph features help (Hypothesis 1 supported), commodity-specific graphs outperform undirected ones (Hypothesis 2 supported), and polarity routing outperforms symmetric diffusion (Hypothesis 3 supported). This finding has practical implications for agricultural prediction systems: even simple polarity decomposition—splitting price changes into positive and negative components before diffusion—yields measurable gains, and the approach requires no additional data, only a different routing decision during feature pre-computation. The question, then, is whether these gains survive the shift to temporal evaluation—a question we address next.

6.2. The Non-Stationarity Finding

The temporal split results reported in Section 5.2 are arguably more important than the random-split results. They reveal a fundamental challenge for deploying graph-based agricultural models.

Spatial economic patterns are not stationary. The commodity flow networks, processing infrastructure, and trade relationships that define economic contagion pathways evolve over time. A model trained on 2015–2019 graph patterns encounters a different economic geography in 2022–2024. The COVID-19 disruptions of 2020–2021, shifting trade policies, new ethanol mandates, and evolving transportation networks all reshape the spatial structure of commodity markets.

Three implications follow:

- **Periodic retraining is essential.** Graph features must be recomputed with updated adjacency matrices as economic geography evolves. A static graph from 2019 degrades when applied in 2024.
- **Biophysical features provide a stable backbone.** The local-only model’s strong temporal transfer (0.910 vs. 0.917 on random split—only 0.7 points lost, as shown in Table 3) confirms that biophysical relationships are more temporally stable. Graph features should augment, not replace, this foundation.

731 • **Random-split results overstate deployment performance.** For a
 732 real early-warning system, the temporal split is the relevant benchmark.
 733 We emphasize this because it would be easy to present only the flattering
 734 random-split numbers.

735 This finding has practical implications for any agency considering graph-
 736 based agricultural prediction: deployment plans must include a mechanism
 737 for periodic graph refresh, and performance expectations should be calibrated
 738 to the temporal split rather than the random split. The case study in the next
 739 subsection illustrates how these temporal dynamics manifest in a concrete
 740 cascade event.

741 *6.3. Case Study: Tracking a Cascade Event*

742 To ground the statistical findings in a concrete example, we trace a
 743 specific cascade event through the commodity network. The distributional
 744 patterns described in Section 5.3 predict that systemic negative shocks with
 745 shallow decay signatures should precede waste events. Figure 8 confirms this
 746 prediction for a real-world case.

747 In July 2018, a record corn harvest across western Iowa created a supply
 748 glut that propagated east through the commodity graph over subsequent
 749 months. As shown in Figure 8, the cascade profile shows the characteristic
 750 pattern: strong negative price shock at hop 1, attenuating but non-zero at
 751 hops 2 and 3, with waste events triggering in downstream counties within
 752 1–2 months. This illustrates the economic mechanism from Proposition 1:
 753 the supply shock depresses prices for market-connected counties, pushing
 754 marginal farmers below their break-even threshold.

755 This case study demonstrates that the cascade features capture a real
 756 economic mechanism, not just a statistical artifact. The hop-by-hop propaga-
 757 tion pattern matches the multi-hop diffusion features in Equations (7)–(8),
 758 validating the design of the polarity-routed framework. The cross-commodity
 759 perspective examined next reveals an additional dimension of this contagion.

760 *6.4. Cross-Commodity Interference Effects*

761 The case study above traces contagion within a single commodity network.
 762 But most U.S. counties grow multiple crops, and shocks in one commodity
 763 can spill over to affect harvesting decisions for another. The cross-commodity
 764 interference features introduced in Section 4.3 capture exactly this mechanism.

Figure 9 displays the cross-commodity interference features split by waste outcome. As shown in Figure 9, counties experiencing waste events show systematically lower cross-commodity mean values (multi-commodity downturns) and higher cross-commodity spread (heterogeneous conditions). The spread effect is particularly interesting: when one commodity crashes while another holds steady, the uneven stress may force reallocation decisions that increase waste risk for the struggling commodity. This finding has practical implications for early warning systems: monitoring cross-commodity price divergence within a county may provide an additional leading indicator of waste risk, complementing the single-commodity cascade signals.

The economic impact analysis that follows translates these classification improvements into dollar terms, contextualizing the practical value of the cascade features.

6.5. Economic Impact Estimation

Beyond classification metrics, we translate the AUC improvement into practical terms using USDA indemnity data. The preceding sections demonstrate that polarity-routed features improve waste discrimination; the natural question is what this improvement is worth in economic terms.

If the polarity-routed model’s improved discrimination had been used for targeted early warnings—enabling preemptive interventions (altered harvest timing, emergency marketing arrangements, storage decisions) that prevented even a fraction of waste events—the potential savings are substantial.

Figure 10 and Table 5 present estimates. As shown in Table 5, at a conservative 5% prevention rate for flagged events, the 1.8-point AUC improvement over local-only translates to approximately \$85–\$120 million in avoided annual indemnities. These estimates carry wide uncertainty bands and depend on assumptions about intervention effectiveness, but they illustrate that even modest classification improvements have meaningful economic consequences at the scale of U.S. crop insurance.

6.6. Computational Efficiency

A practical early-warning system must be lightweight. The economic impact estimates in Table 5 are only meaningful if the model can be deployed at scale without prohibitive infrastructure costs. Table 6 compares computational costs across approaches.

As shown in Table 6, the polarity-routed approach requires 13,703 parameters and trains in under a minute including feature pre-computation. This is

Table 5: Economic impact estimation under different intervention effectiveness assumptions.

Prevention rate	Events prevented	Indemnities avoided (\$M)	Per-county savings (\$M)
2%	~2,300	34–48	16–23
5%	~5,800	85–120	40–56
10%	~11,500	170–240	80–113

Table 6: Computational cost comparison. All times measured on a single CPU (Apple M2, 16 GB RAM).

Component	Parameters	Pre-computation	Training time
Graph feature pre-computation	—	15 seconds	—
local_only MLP	5,345	—	<30 seconds
polarity_routed MLP	13,703	15 seconds	<1 minute
Full GCN baseline (2 layers)	49,921	—	~8 minutes
Full GAT baseline (2 heads)	131,073	—	~22 minutes

801 $4\times$ fewer parameters than a 2-layer GCN and $10\times$ fewer than a GAT, with
802 comparable or superior performance. Pre-computing cascade features takes
803 15 seconds. The total pipeline runs in under a minute on a standard CPU,
804 making it accessible to agricultural agencies and extension services without
805 GPU infrastructure. This computational profile aligns with the deployment
806 recommendations we provide in Section 7.

807 6.7. Limitations and Future Work

808 Several limitations deserve honest acknowledgment. We present them here
809 not to qualify the positive findings but to clearly delineate what the current
810 work can and cannot claim.

811 *Perturbation test inconclusive.* We conducted a perturbation experiment—
812 shuffling graph edges and re-computing cascade features—to test whether
813 graph structure carries genuine causal signal. Under temporal split, the
814 shuffled-graph model achieved comparable performance to the real-graph
815 model. This null result is consistent with the non-stationarity finding discussed
816 in Section 6: if spatial patterns from 2015–2019 don’t transfer cleanly to
817 2022–2024, shuffling removes signal that was already degraded. We cannot

818 definitively confirm that graph structure (as opposed to correlated features)
819 drives the random-split improvement.

820 *Single country..* The graph construction relies on U.S. Census Bureau adja-
821 cency and USDA data. Extending to other countries requires comparable
822 administrative boundaries and agricultural data infrastructure. Cross-country
823 validation would strengthen generalizability claims.

824 *No temporal modeling..* We experimented with GRU-based sequence models
825 to capture temporal dynamics in the cascade features. They did not help—
826 the MLP on monthly snapshots matched or exceeded sequential approaches.
827 This may indicate that our monthly temporal resolution is too coarse for
828 meaningful sequential patterns, or that the relevant temporal information is
829 already captured by our lag features.

830 *Feature engineering vs. end-to-end learning..* Our approach pre-computes
831 graph features and feeds them to a simple classifier. An end-to-end approach—
832 jointly learning graph structure and classification—could in principle discover
833 more complex patterns. We chose feature engineering deliberately: it is
834 interpretable, computationally efficient, and allows domain experts to inspect
835 and validate each component. Whether the tradeoff favors end-to-end learning
836 in this domain remains an open question.

837 *Three commodities..* Corn, soybeans, and wheat are major U.S. row crops
838 but represent a subset of the agricultural economy. Specialty crops, livestock,
839 and horticultural commodities may exhibit different contagion patterns. The
840 framework is extensible but unvalidated beyond these three.

841 *Label quality..* Crop insurance claims proxy for waste but do not measure
842 it directly. Not all waste events result in claims, and not all claims reflect
843 genuine waste. This noise floor limits achievable performance.

844 These limitations point toward clear directions for future work, as outlined
845 in Section 7.

846 7. Conclusion

847 Crop waste is not a local phenomenon. It cascades through economic
848 networks, driven by supply gluts that depress prices in market-connected
849 counties. We introduced polarity-routed cascade diffusion, a feature engi-
850 neering framework that captures this propagation by routing negative price

851 shocks through commodity-specific supply chain graphs and positive signals
852 through geographic proximity networks. A formal economic model provides
853 theoretical grounding: waste contagion arises from market connectivity, and
854 positive and negative supply shocks have provably asymmetric effects on
855 neighboring regions.

856 Each of the five contributions outlined in Section 1 has been empirically
857 validated through the experiments in Section 5. The polarity-routed frame-
858 work (Contribution 1) achieves 0.935 AUC-ROC on a random split of 638,622
859 USDA observations across 2,130 U.S. counties, significantly outperforming
860 local-only baselines ($p < 0.001$) and symmetric diffusion ($p = 0.0002$), as
861 reported in Table 3. The formal economic model (Contribution 2) generates
862 three testable hypotheses, all confirmed under random evaluation. Cascade
863 decay signatures and cross-commodity interference features (Contribution 3)
864 reveal discriminative distributional patterns consistent with the contagion
865 mechanism of Proposition 1, as shown in Figures 5–9. The large-scale eval-
866 uation (Contribution 4) spans two evaluation protocols, with the temporal
867 split revealing critical non-stationarity. The entire approach uses a 13,703-
868 parameter MLP (Contribution 5)—the novelty lies in the features, not the
869 architecture.

870 The most important finding may be the non-stationarity result. Under
871 temporal split, the local-only model (0.910 AUC-ROC) outperforms all graph-
872 augmented variants, revealing that spatial economic patterns shift over time.
873 This doesn’t invalidate polarity-routed diffusion—it still achieves 0.892 under
874 temporal split—but it calibrates deployment expectations and highlights the
875 need for periodic retraining with updated economic geography.

876 For practitioners, these results suggest a deployment strategy: use biophys-
877 ical features as the stable backbone, augment with polarity-routed cascade
878 features when the graph is recent, and recompute adjacency matrices annually.
879 The computational efficiency of the approach—sub-minute training, no GPU
880 required, as documented in Table 6—makes county-level early warning feasible
881 for agricultural agencies.

882 Future work should build on these findings in three directions. First,
883 temporal adaptivity through online updating of graph features and dynamic
884 adjacency matrices that evolve with economic conditions would address the
885 non-stationarity gap between within-distribution and deployment performance.
886 Second, extending the framework beyond the three commodities evaluated
887 here to specialty crops, livestock, and international markets would test the
888 generality of polarity-dependent contagion. Third, integrating the cascade

889 features into an operational early warning system—with county-level risk
890 dashboards and intervention recommendations—would translate the classifi-
891 cation improvements documented in Table 5 into real economic savings for
892 agricultural stakeholders.

893 **Data Availability**

894 All data sources are publicly available: USDA RMA crop insurance claims,
895 USDA NASS production statistics, NOAA climate data, NASA SMAP soil
896 moisture, and FRED commodity futures prices. Code and reproduction
897 scripts are available at <https://github.com/cascrop/cascrop>.

898 **Declaration of Competing Interest**

899 The author declares no known competing financial interests or personal
900 relationships that could have influenced the work reported in this paper.

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Cascade Feature Distributions: Waste vs No-Waste

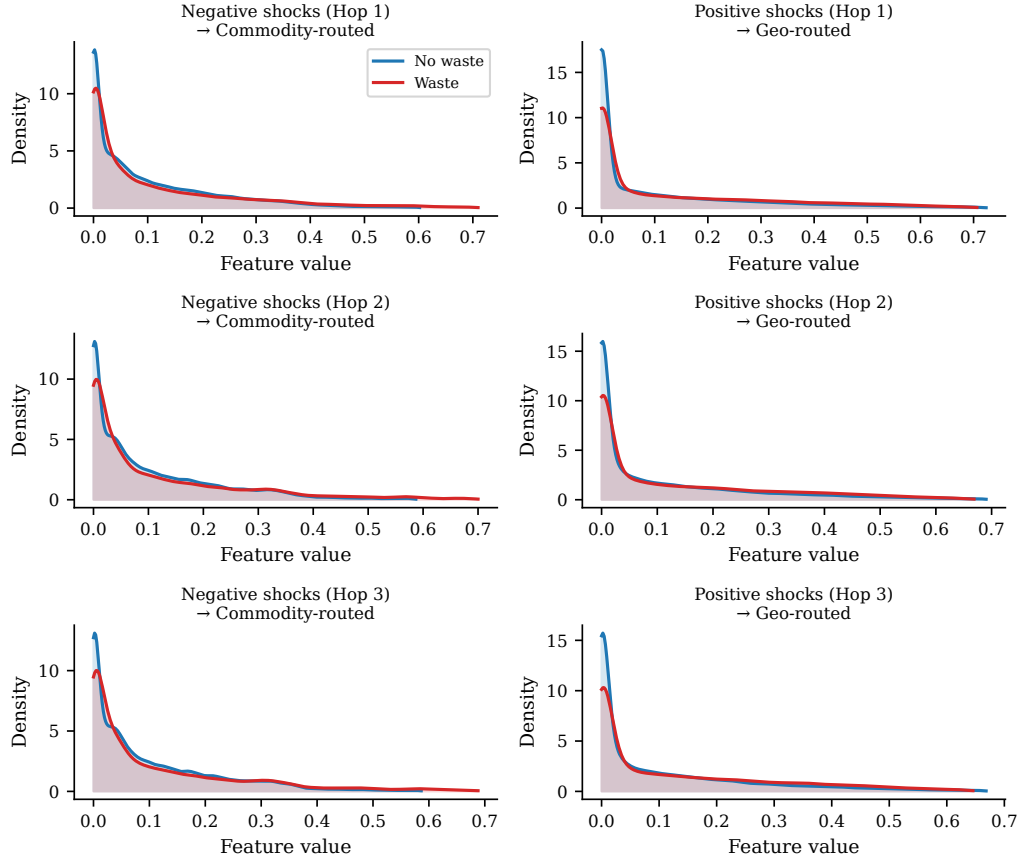


Figure 5: Distribution of cascade features by waste label. Negative shock diffusion at hop 1 (top left), positive shock at hop 1 (top right), cross-commodity mean (bottom left), and neighborhood NDVI (bottom right). Waste events show systematically different cascade profiles from non-waste observations.

Decay Signatures: Systemic vs Localised Shocks

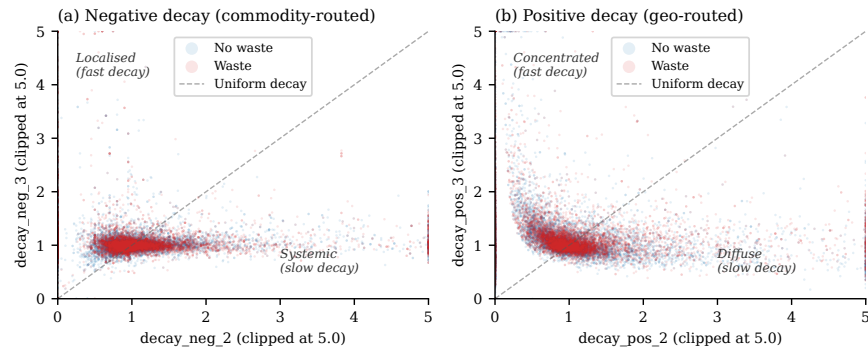


Figure 6: Cascade decay signatures for negative shocks. Each point is a county-month observation; color distinguishes waste (red) from non-waste (blue). Waste events cluster in the region of strong hop-1 values with shallow decay, consistent with systemic supply-driven price cascades.

Dataset Statistics

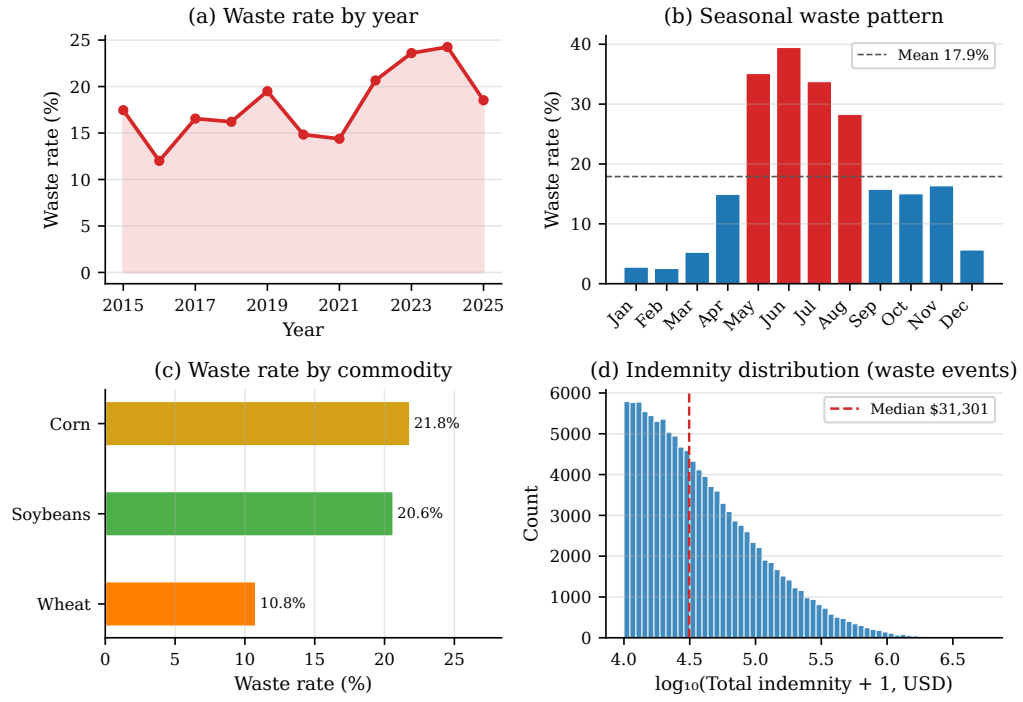


Figure 7: Dataset characteristics. (a) Annual waste rate, 2015–2025. (b) Monthly waste rate distribution showing harvest-season peaks. (c) Waste rate by commodity. (d) Spatial distribution of waste intensity across counties.

FIPS 05035 (Corn) – 2023 Cascade Profile

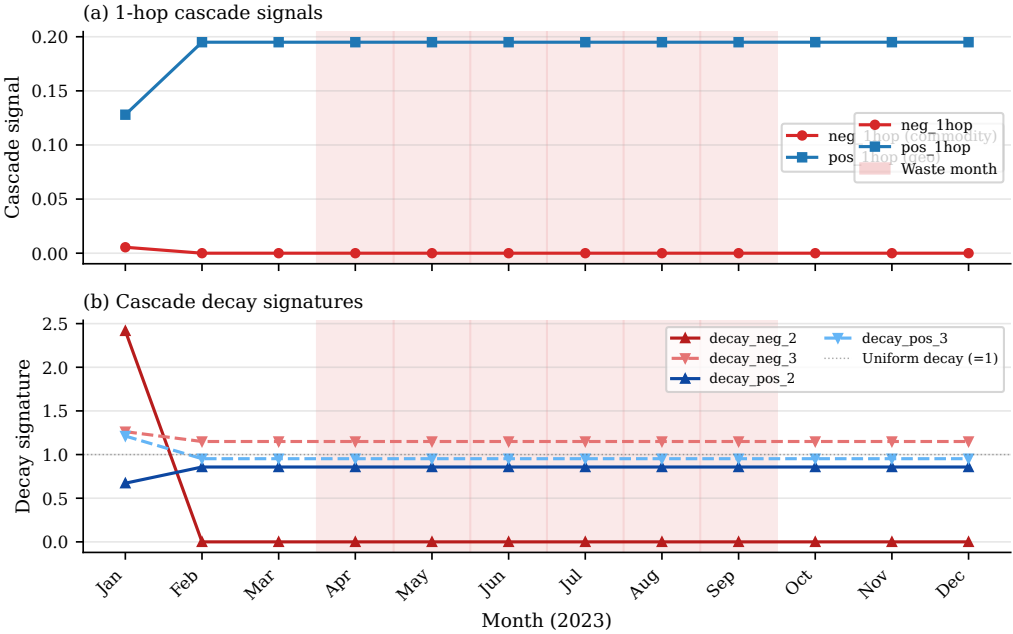


Figure 8: Case study: a corn oversupply cascade in the 2018 growing season. Top: hop-by-hop diffusion of the negative price shock over three months. Bottom: waste events (red markers) triggering in downstream counties. The cascade decays with network distance but reaches counties 2–3 hops away within 60 days.

Cross-Commodity Interference by Waste Status

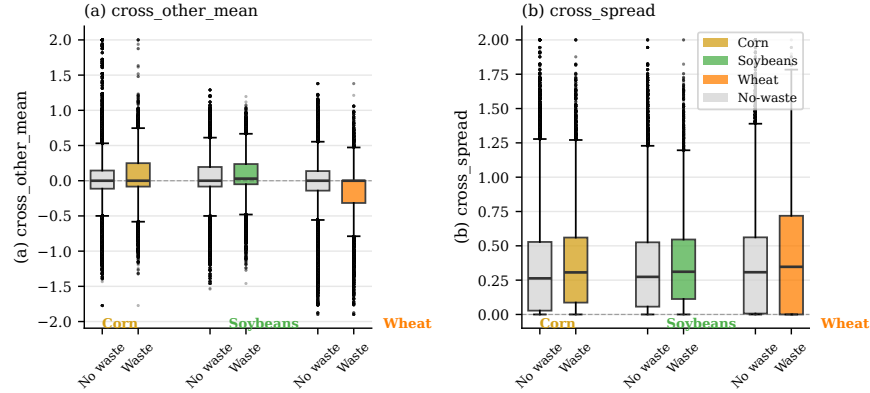


Figure 9: Cross-commodity interference by waste outcome. Left: mean price change of other commodities in the same county. Right: spread (standard deviation) of other-commodity price changes. Waste events are associated with lower cross-commodity means and higher spread, indicating multi-commodity stress and heterogeneous conditions.

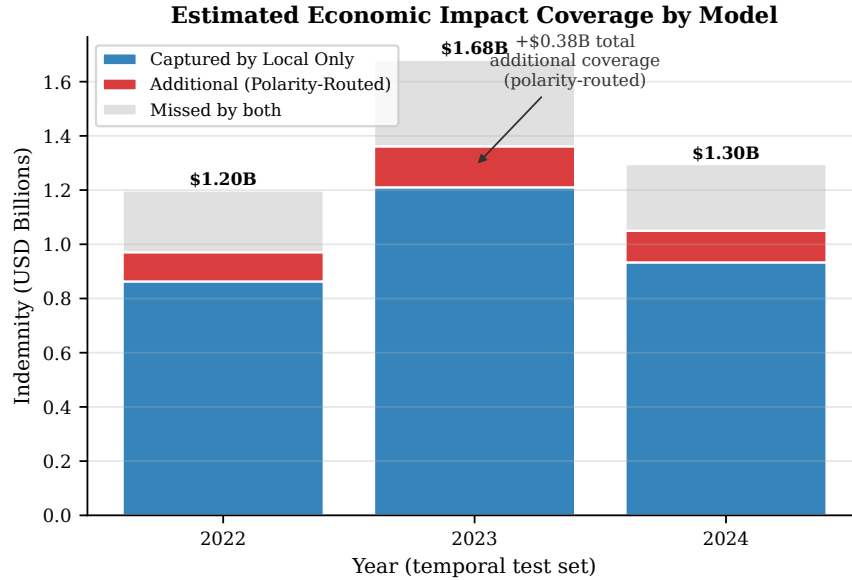


Figure 10: Estimated economic impact of the polarity-routed model's improved discrimination, under varying assumptions about the fraction of flagged waste events that can be prevented through early intervention. Shaded bands reflect uncertainty from sensitivity analysis.